Tecnológico De Estudios Superiores De Villa Guerrero

Actividad: Practica 1

EXPLORANDO PROCESOS CON PS EN LINUX

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Carrera: Ingeniería en Sistemas Computacionales

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Grupo: 3501

Introducción

En esta práctica se exploró el comando ps en un sistema operativo Linux. Este comando permite visualizar información relacionada con los procesos que se encuentran activos en el sistema.

Se utilizaron diferentes opciones de ps para observar los procesos del usuario actual, consultar información detallada, filtrar procesos, consultar los procesos de un usuario específico, mostrar el PID y el nombre de los comandos, así como ordenar los procesos de acuerdo con su consumo de CPU.

Objetivo

Aprender a utilizar el comando ps para visualizar, filtrar y comprender los procesos activos en un sistema Linux.

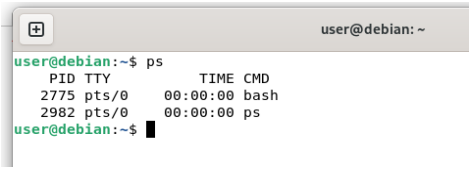
Desarrollo de la práctica

Para comenzar se abrió la terminal



Despues se ejecuto el comando

ps



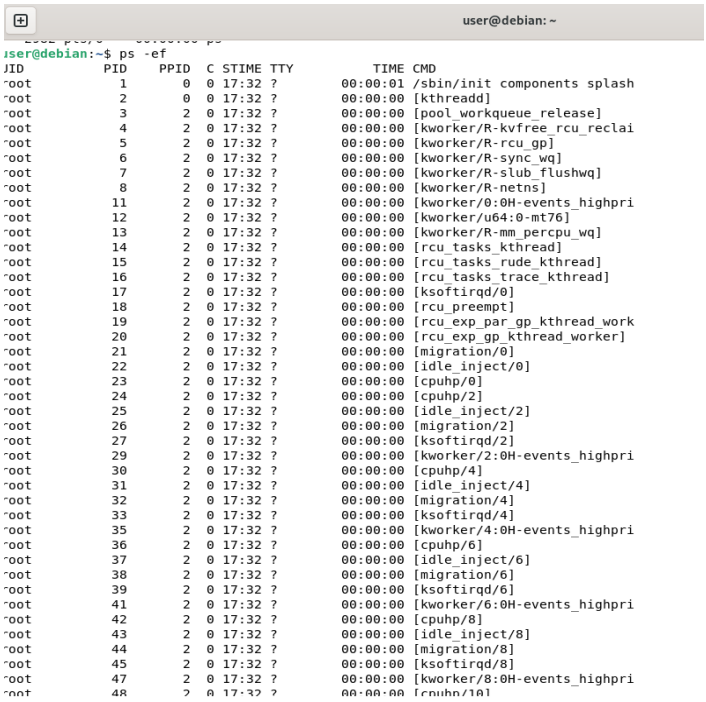
Este comando permite visualizar los procesos asociados al usuario actual.

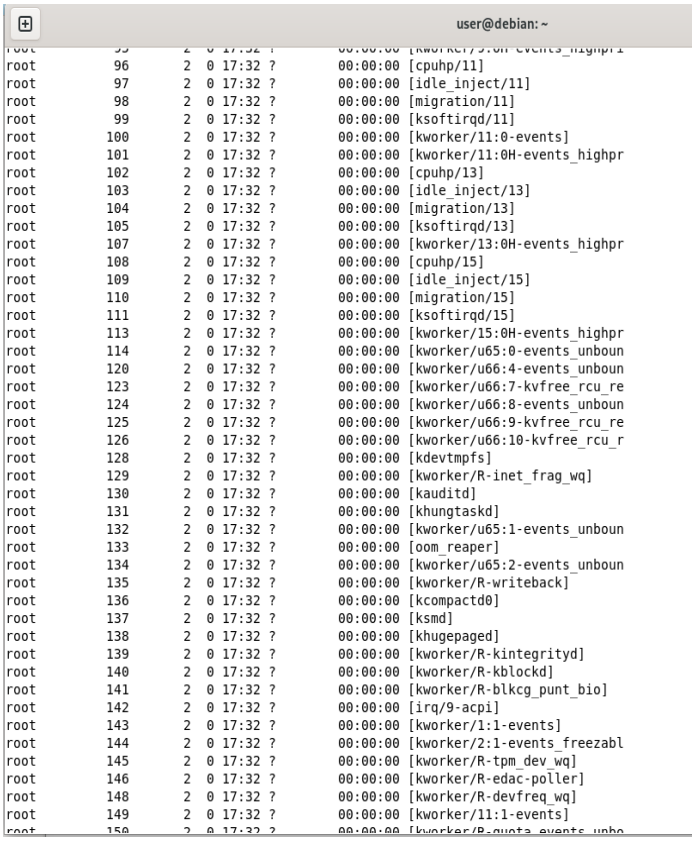
Resultado: Después de ejecutar el comando se mostró una lista de los procesos que se encontraban activos para el usuario. En la información mostrada se pueden observar datos como el PID y el comando correspondiente.

Posteriormente se ejecutó:

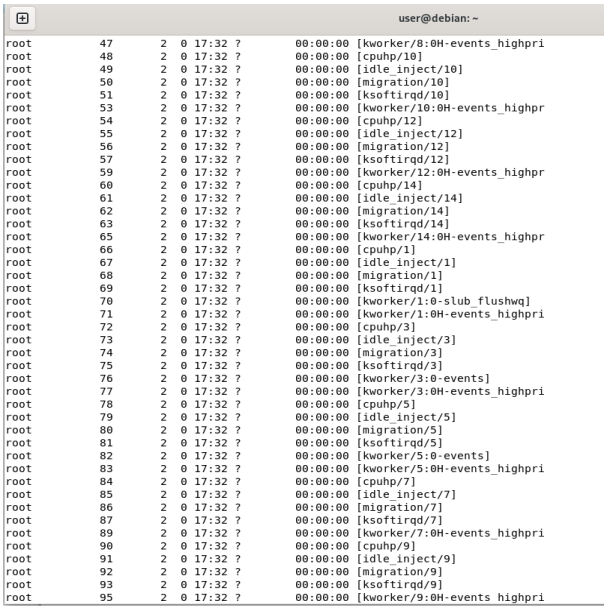
ps -ef

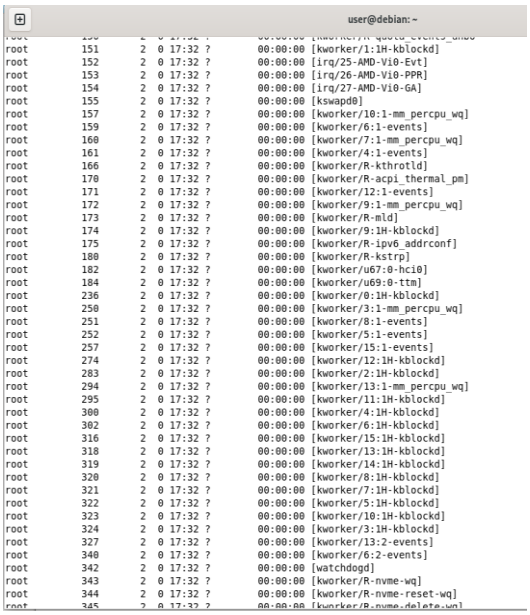
La opción -ef permite mostrar información más detallada de los procesos que se encuentran ejecutándose en el sistema.

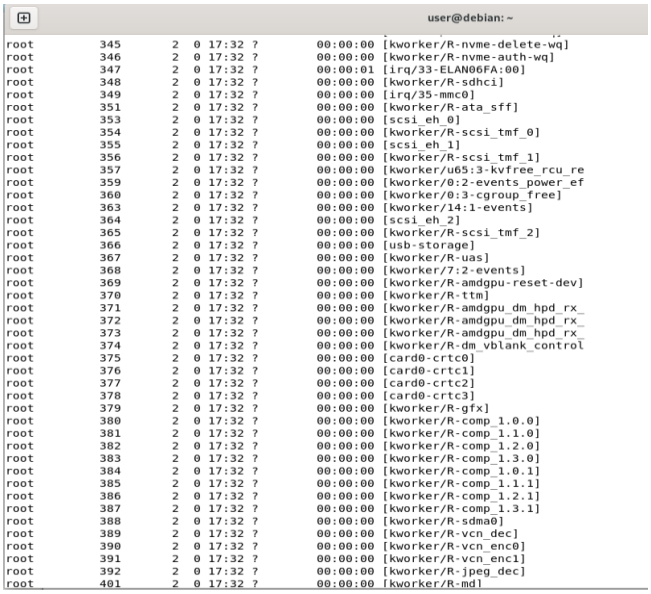


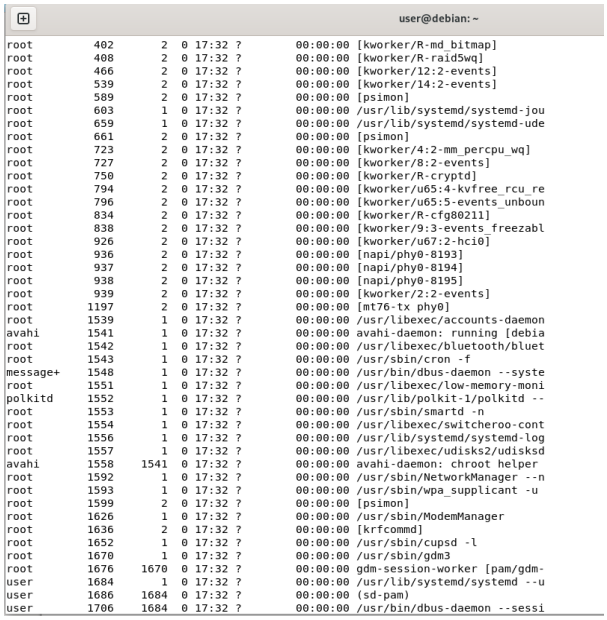


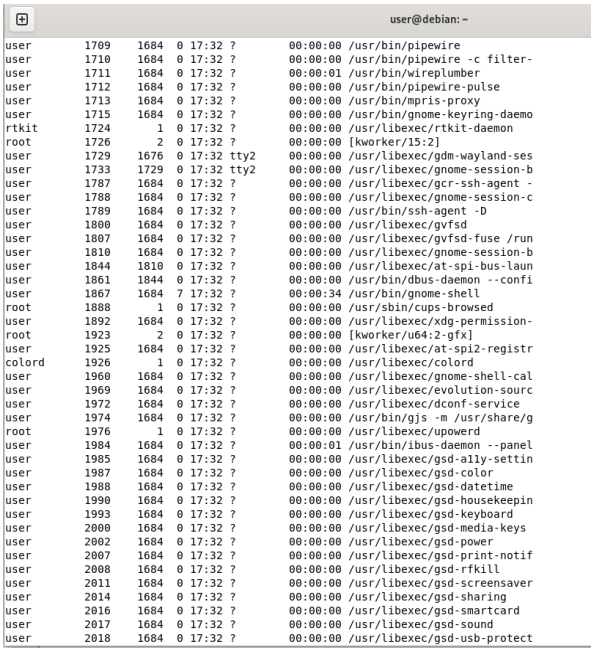
Resultado: Se obtuvo una lista más amplia de procesos junto con información adicional, como el usuario que ejecuta el proceso, el PID y otros datos relacionados con su ejecución.


user@debian: ~
root 47 2 0 17:32 ? 00:00:00 [kworker/8:0H-events_highpri
root 48 2 0 17:32 ? 00:00:00 [cpuhp/10]
root 49 2 0 17:32 ? 00:00:00 [idle_inject/10]
root 50 2 0 17:32 ? 00:00:00 [migration/10]
root 51 2 0 17:32 ? 00:00:00 [ksoftirqd/10]
root 53 2 0 17:32 ? 00:00:00 [kworker/10:0H-events_highpr
root 54 2 0 17:32 ? 00:00:00 [cpuhp/12]
root 55 2 0 17:32 ? 00:00:00 [idle_inject/12]
root 56 2 0 17:32 ? 00:00:00 [migration/12]
root 57 2 0 17:32 ? 00:00:00 [ksoftirqd/12]
root 59 2 0 17:32 ? 00:00:00 [kworker/12:0H-events_highpr
root 60 2 0 17:32 ? 00:00:00 [cpuhp/14]
root 61 2 0 17:32 ? 00:00:00 [idle_inject/14]
root 62 2 0 17:32 ? 00:00:00 [migration/14]
root 63 2 0 17:32 ? 00:00:00 [ksoftirqd/14]
root 65 2 0 17:32 ? 00:00:00 [kworker/14:0H-events_highpr
root 66 2 0 17:32 ? 00:00:00 [cpuhp/1]
root 67 2 0 17:32 ? 00:00:00 [idle_inject/1]
root 68 2 0 17:32 ? 00:00:00 [migration/1]
root 69 2 0 17:32 ? 00:00:00 [ksoftirqd/1]
root 70 2 0 17:32 ? 00:00:00 [kworker/1:0-slub_flushwq]
root 71 2 0 17:32 ? 00:00:00 [kworker/1:0H-events_highpri
root 72 2 0 17:32 ? 00:00:00 [cpuhp/3]
root 73 2 0 17:32 ? 00:00:00 [idle_inject/3]
root 74 2 0 17:32 ? 00:00:00 [migration/3]
root 75 2 0 17:32 ? 00:00:00 [ksoftirqd/3]
root 76 2 0 17:32 ? 00:00:00 [kworker/3:0-events]
root 77 2 0 17:32 ? 00:00:00 [kworker/3:0H-events_highpri
root 78 2 0 17:32 ? 00:00:00 [cpuhp/5]
root 79 2 0 17:32 ? 00:00:00 [idle_inject/5]
root 80 2 0 17:32 ? 00:00:00 [migration/5]
root 81 2 0 17:32 ? 00:00:00 [ksoftirqd/5]
root 82 2 0 17:32 ? 00:00:00 [kworker/5:0-events]
root 83 2 0 17:32 ? 00:00:00 [kworker/5:0H-events_highpri
root 84 2 0 17:32 ? 00:00:00 [cpuhp/7]
root 85 2 0 17:32 ? 00:00:00 [idle_inject/7]
root 86 2 0 17:32 ? 00:00:00 [migration/7]
root 87 2 0 17:32 ? 00:00:00 [ksoftirqd/7]
root 89 2 0 17:32 ? 00:00:00 [kworker/7:0H-events_highpri
root 90 2 0 17:32 ? 00:00:00 [cpuhp/9]
root 91 2 0 17:32 ? 00:00:00 [idle_inject/9]
root 92 2 0 17:32 ? 00:00:00 [migration/9]
root 93 2 0 17:32 ? 00:00:00 [ksoftirqd/9]
root 95 2 0 17:32 ? 00:00:00 [kworker/9:0H-events_highpri

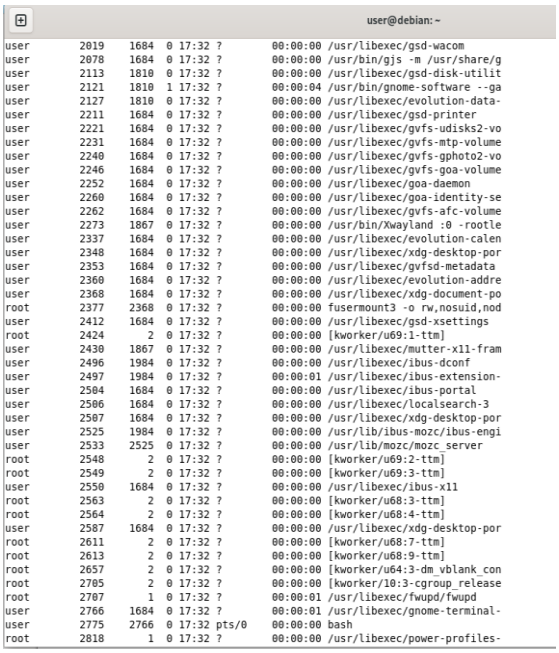

user@debian: ~
root 151 2 0 17:32 ? 00:00:00 [kworker/1:1H-kblockd]
root 152 2 0 17:32 ? 00:00:00 [irq/25-AMD-Vi0-Evt]
root 153 2 0 17:32 ? 00:00:00 [irq/26-AMD-Vi0-PPR]
root 154 2 0 17:32 ? 00:00:00 [irq/27-AMD-Vi0-GA]
root 155 2 0 17:32 ? 00:00:00 [kswapd0]
root 157 2 0 17:32 ? 00:00:00 [kworker/10:1-mm_percpu_wq]
root 159 2 0 17:32 ? 00:00:00 [kworker/6:1-events]
root 160 2 0 17:32 ? 00:00:00 [kworker/7:1-mm_percpu_wq]
root 161 2 0 17:32 ? 00:00:00 [kworker/4:1-events]
root 166 2 0 17:32 ? 00:00:00 [kworker/R-kthrotld]
root 170 2 0 17:32 ? 00:00:00 [kworker/R-acpi_thermal_pm]
root 171 2 0 17:32 ? 00:00:00 [kworker/12:1-events]
root 172 2 0 17:32 ? 00:00:00 [kworker/9:1-mm_percpu_wq]
root 173 2 0 17:32 ? 00:00:00 [kworker/R-mld]
root 174 2 0 17:32 ? 00:00:00 [kworker/9:1H-kblockd]
root 175 2 0 17:32 ? 00:00:00 [kworker/R-ipv6_addrconf]
root 180 2 0 17:32 ? 00:00:00 [kworker/R-kstrp]
root 182 2 0 17:32 ? 00:00:00 [kworker/u67:0-hci0]
root 184 2 0 17:32 ? 00:00:00 [kworker/u69:0-ttm]
root 236 2 0 17:32 ? 00:00:00 [kworker/0:1H-kblockd]
root 250 2 0 17:32 ? 00:00:00 [kworker/3:1-mm_percpu_wq]
root 251 2 0 17:32 ? 00:00:00 [kworker/8:1-events]
root 252 2 0 17:32 ? 00:00:00 [kworker/5:1-events]
root 257 2 0 17:32 ? 00:00:00 [kworker/15:1-events]
root 274 2 0 17:32 ? 00:00:00 [kworker/12:1H-kblockd]
root 283 2 0 17:32 ? 00:00:00 [kworker/2:1H-kblockd]
root 294 2 0 17:32 ? 00:00:00 [kworker/13:1-mm_percpu_wq]
root 295 2 0 17:32 ? 00:00:00 [kworker/11:1H-kblockd]
root 300 2 0 17:32 ? 00:00:00 [kworker/4:1H-kblockd]
root 302 2 0 17:32 ? 00:00:00 [kworker/6:1H-kblockd]
root 316 2 0 17:32 ? 00:00:00 [kworker/15:1H-kblockd]
root 318 2 0 17:32 ? 00:00:00 [kworker/13:1H-kblockd]
root 319 2 0 17:32 ? 00:00:00 [kworker/14:1H-kblockd]
root 320 2 0 17:32 ? 00:00:00 [kworker/8:1H-kblockd]
root 321 2 0 17:32 ? 00:00:00 [kworker/7:1H-kblockd]
root 322 2 0 17:32 ? 00:00:00 [kworker/5:1H-kblockd]
root 323 2 0 17:32 ? 00:00:00 [kworker/10:1H-kblockd]
root 324 2 0 17:32 ? 00:00:00 [kworker/3:1H-kblockd]
root 327 2 0 17:32 ? 00:00:00 [kworker/13:2-events]
root 340 2 0 17:32 ? 00:00:00 [kworker/6:2-events]
root 342 2 0 17:32 ? 00:00:00 [watchdogd]
root 343 2 0 17:32 ? 00:00:00 [kworker/R-nvme-wq]
root 344 2 0 17:32 ? 00:00:00 [kworker/R-nvme-reset-wq]


user@debian: ~
root 345 2 0 17:32 ? 00:00:00 [kworker/R-nvme-delete-wq]
root 346 2 0 17:32 ? 00:00:00 [kworker/R-nvme-auth-wq]
root 347 2 0 17:32 ? 00:00:01 [irq/33-ELAN06FA:00]
root 348 2 0 17:32 ? 00:00:00 [kworker/R-sdhci]
root 349 2 0 17:32 ? 00:00:00 [irq/35-mmc0]
root 351 2 0 17:32 ? 00:00:00 [kworker/R-ata_sff]
root 353 2 0 17:32 ? 00:00:00 [scsi_eh_0]
root 354 2 0 17:32 ? 00:00:00 [kworker/R-scsi_tmf_0]
root 355 2 0 17:32 ? 00:00:00 [scsi_eh_1]
root 356 2 0 17:32 ? 00:00:00 [kworker/R-scsi_tmf_1]
root 357 2 0 17:32 ? 00:00:00 [kworker/u65:3-kvfree_rcu_re
root 359 2 0 17:32 ? 00:00:00 [kworker/0:2-events_power_ef
root 360 2 0 17:32 ? 00:00:00 [kworker/0:3-cgroup_free]
root 363 2 0 17:32 ? 00:00:00 [kworker/14:1-events]
root 364 2 0 17:32 ? 00:00:00 [scsi_eh_2]
root 365 2 0 17:32 ? 00:00:00 [kworker/R-scsi_tmf_2]
root 366 2 0 17:32 ? 00:00:00 [usb-storage]
root 367 2 0 17:32 ? 00:00:00 [kworker/R-uas]
root 368 2 0 17:32 ? 00:00:00 [kworker/7:2-events]
root 369 2 0 17:32 ? 00:00:00 [kworker/R-amdgpu-reset-dev]
root 370 2 0 17:32 ? 00:00:00 [kworker/R-ttm]
root 371 2 0 17:32 ? 00:00:00 [kworker/R-amdgpu_dm_hpd_rx_
root 372 2 0 17:32 ? 00:00:00 [kworker/R-amdgpu_dm_hpd_rx_
root 373 2 0 17:32 ? 00:00:00 [kworker/R-amdgpu_dm_hpd_rx_
root 374 2 0 17:32 ? 00:00:00 [kworker/R-dm_vblank_control
root 375 2 0 17:32 ? 00:00:00 [card0-crtc0]
root 376 2 0 17:32 ? 00:00:00 [card0-crtc1]
root 377 2 0 17:32 ? 00:00:00 [card0-crtc2]
root 378 2 0 17:32 ? 00:00:00 [card0-crtc3]
root 379 2 0 17:32 ? 00:00:00 [kworker/R-gfx]
root 380 2 0 17:32 ? 00:00:00 [kworker/R-comp_1.0.0]
root 381 2 0 17:32 ? 00:00:00 [kworker/R-comp_1.1.0]
root 382 2 0 17:32 ? 00:00:00 [kworker/R-comp_1.2.0]
root 383 2 0 17:32 ? 00:00:00 [kworker/R-comp_1.3.0]
root 384 2 0 17:32 ? 00:00:00 [kworker/R-comp_1.0.1]
root 385 2 0 17:32 ? 00:00:00 [kworker/R-comp_1.1.1]
root 386 2 0 17:32 ? 00:00:00 [kworker/R-comp_1.2.1]
root 387 2 0 17:32 ? 00:00:00 [kworker/R-comp_1.3.1]
root 388 2 0 17:32 ? 00:00:00 [kworker/R-sdma0]
root 389 2 0 17:32 ? 00:00:00 [kworker/R-vcn_dec]
root 390 2 0 17:32 ? 00:00:00 [kworker/R-vcn_enc0]
root 391 2 0 17:32 ? 00:00:00 [kworker/R-vcn_enc1]
root 392 2 0 17:32 ? 00:00:00 [kworker/R-jpeg_dec]
root 401 2 0 17:32 ? 00:00:00 [kworker/R-md]

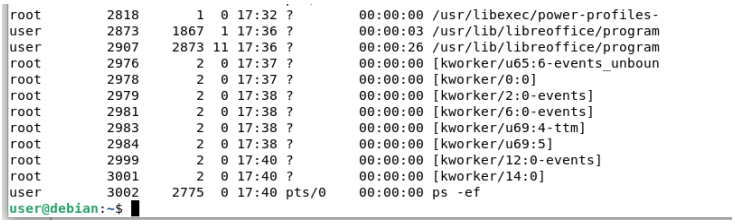

user@debian: ~
root 402 2 0 17:32 ? 00:00:00 [kworker/R-md_bitmap]
root 408 2 0 17:32 ? 00:00:00 [kworker/R-raid5wq]
root 466 2 0 17:32 ? 00:00:00 [kworker/12:2-events]
root 539 2 0 17:32 ? 00:00:00 [kworker/14:2-events]
root 589 2 0 17:32 ? 00:00:00 [psimon]
root 603 1 0 17:32 ? 00:00:00 /usr/lib/systemd/systemd-jou
root 659 1 0 17:32 ? 00:00:00 /usr/lib/systemd/systemd-ude
root 661 2 0 17:32 ? 00:00:00 [psimon]
root 723 2 0 17:32 ? 00:00:00 [kworker/4:2-mm_percpu_wq]
root 727 2 0 17:32 ? 00:00:00 [kworker/8:2-events]
root 750 2 0 17:32 ? 00:00:00 [kworker/R-cryptd]
root 794 2 0 17:32 ? 00:00:00 [kworker/u65:4-kvfree_rcu_re
root 796 2 0 17:32 ? 00:00:00 [kworker/u65:5-events_unboun
root 834 2 0 17:32 ? 00:00:00 [kworker/R-cfg80211]
root 838 2 0 17:32 ? 00:00:00 [kworker/9:3-events_freezabl
root 926 2 0 17:32 ? 00:00:00 [kworker/u67:2-hci0]
root 936 2 0 17:32 ? 00:00:00 [napi/phy0-8193]
root 937 2 0 17:32 ? 00:00:00 [napi/phy0-8194]
root 938 2 0 17:32 ? 00:00:00 [napi/phy0-8195]
root 939 2 0 17:32 ? 00:00:00 [kworker/2:2-events]
root 1197 2 0 17:32 ? 00:00:00 [mt76-tx phy0]
root 1539 1 0 17:32 ? 00:00:00 /usr/libexec/accounts-daemon
avahi 1541 1 0 17:32 ? 00:00:00 avahi-daemon: running [debia
root 1542 1 0 17:32 ? 00:00:00 /usr/libexec/bluetooth/bluet
root 1543 1 0 17:32 ? 00:00:00 /usr/sbin/cron -f
message+ 1548 1 0 17:32 ? 00:00:00 /usr/bin/dbus-daemon --syste
root 1551 1 0 17:32 ? 00:00:00 /usr/libexec/low-memory-moni
polkitd 1552 1 0 17:32 ? 00:00:00 /usr/lib/polkit-1/polkitd --
root 1553 1 0 17:32 ? 00:00:00 /usr/sbin/smartd -n
root 1554 1 0 17:32 ? 00:00:00 /usr/libexec/switcheroo-cont
root 1556 1 0 17:32 ? 00:00:00 /usr/lib/systemd/systemd-log
root 1557 1 0 17:32 ? 00:00:00 /usr/libexec/udisks2/udisksd
avahi 1558 1541 0 17:32 ? 00:00:00 avahi-daemon: chroot helper
root 1592 1 0 17:32 ? 00:00:00 /usr/sbin/NetworkManager --n
root 1593 1 0 17:32 ? 00:00:00 /usr/sbin/wpa_supplicant -u
root 1599 2 0 17:32 ? 00:00:00 [psimon]
root 1626 1 0 17:32 ? 00:00:00 /usr/sbin/ModemManager
root 1636 2 0 17:32 ? 00:00:00 [krfcommd]
root 1652 1 0 17:32 ? 00:00:00 /usr/sbin/cupsd -l
root 1670 1 0 17:32 ? 00:00:00 /usr/sbin/gdm3
root 1676 1670 0 17:32 ? 00:00:00 gdm-session-worker [pam/gdm-
user 1684 1 0 17:32 ? 00:00:00 /usr/lib/systemd/systemd --u
user 1686 1684 0 17:32 ? 00:00:00 (sd-pam)
user 1706 1684 0 17:32 ? 00:00:00 /usr/bin/dbus-daemon --sessi



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user@debian: ~
user        1709    1684  0 17:32 ?        00:00:00 /usr/bin/pipewire
user        1710    1684  0 17:32 ?        00:00:00 /usr/bin/pipewire -c filter-
user        1711    1684  0 17:32 ?        00:00:01 /usr/bin/wireplumber
user        1712    1684  0 17:32 ?        00:00:00 /usr/bin/pipewire-pulse
user        1713    1684  0 17:32 ?        00:00:00 /usr/bin/mpris-proxy
user        1715    1684  0 17:32 ?        00:00:00 /usr/bin/gnome-keyring-daemo
rtkit       1724       1  0 17:32 ?        00:00:00 /usr/libexec/rtkit-daemon
root        1726       2  0 17:32 ?        00:00:00 [kworker/15:2]
user        1729    1676  0 17:32 tty2     00:00:00 /usr/libexec/gdm-wayland-ses
user        1733    1729  0 17:32 tty2     00:00:00 /usr/libexec/gnome-session-b
user        1787    1684  0 17:32 ?        00:00:00 /usr/libexec/gcr-ssh-agent -
user        1788    1684  0 17:32 ?        00:00:00 /usr/libexec/gnome-session-c
user        1789    1684  0 17:32 ?        00:00:00 /usr/bin/ssh-agent -D
user        1800    1684  0 17:32 ?        00:00:00 /usr/libexec/gvfsd
user        1807    1684  0 17:32 ?        00:00:00 /usr/libexec/gvfsd-fuse /run
user        1810    1684  0 17:32 ?        00:00:00 /usr/libexec/gnome-session-b
user        1844    1810  0 17:32 ?        00:00:00 /usr/libexec/at-spi-bus-laun
user        1861    1844  0 17:32 ?        00:00:00 /usr/bin/dbus-daemon --confi
user        1867    1684  7 17:32 ?        00:00:34 /usr/bin/gnome-shell
root        1888       1  0 17:32 ?        00:00:00 /usr/sbin/cups-browsed
user        1892    1684  0 17:32 ?        00:00:00 /usr/libexec/xdg-permission-
root        1923       2  0 17:32 ?        00:00:00 [kworker/u64:2-gfx]
user        1925    1684  0 17:32 ?        00:00:00 /usr/libexec/at-spi2-registr
colord      1926       1  0 17:32 ?        00:00:00 /usr/libexec/colord
user        1960    1684  0 17:32 ?        00:00:00 /usr/libexec/gnome-shell-cal
user        1969    1684  0 17:32 ?        00:00:00 /usr/libexec/evolution-sourc
user        1972    1684  0 17:32 ?        00:00:00 /usr/libexec/dconf-service
user        1974    1684  0 17:32 ?        00:00:00 /usr/bin/gjs -m /usr/share/g
root        1976       1  0 17:32 ?        00:00:00 /usr/libexec/upowerd
user        1984    1684  0 17:32 ?        00:00:01 /usr/bin/ibus-daemon --panel
user        1985    1684  0 17:32 ?        00:00:00 /usr/libexec/gsd-a11y-settin
user        1987    1684  0 17:32 ?        00:00:00 /usr/libexec/gsd-color
user        1988    1684  0 17:32 ?        00:00:00 /usr/libexec/gsd-datetime
user        1990    1684  0 17:32 ?        00:00:00 /usr/libexec/gsd-housekeepin
user        1993    1684  0 17:32 ?        00:00:00 /usr/libexec/gsd-keyboard
user        2000    1684  0 17:32 ?        00:00:00 /usr/libexec/gsd-media-keys
user        2002    1684  0 17:32 ?        00:00:00 /usr/libexec/gsd-power
user        2007    1684  0 17:32 ?        00:00:00 /usr/libexec/gsd-print-notif
user        2008    1684  0 17:32 ?        00:00:00 /usr/libexec/gsd-rfkill
user        2011    1684  0 17:32 ?        00:00:00 /usr/libexec/gsd-screensaver
user        2014    1684  0 17:32 ?        00:00:00 /usr/libexec/gsd-sharing
user        2016    1684  0 17:32 ?        00:00:00 /usr/libexec/gsd-smartcard
user        2017    1684  0 17:32 ?        00:00:00 /usr/libexec/gsd-sound
user        2018    1684  0 17:32 ?        00:00:00 /usr/libexec/gsd-usb-protect
```



```
user@debian: ~
user        2019    1684  0 17:32 ?        00:00:00 /usr/libexec/gsd-wacom
user        2078    1684  0 17:32 ?        00:00:00 /usr/bin/gjs -m /usr/share/g
user        2113    1810  0 17:32 ?        00:00:00 /usr/libexec/gsd-disk-utilit
user        2121    1810  1 17:32 ?        00:00:04 /usr/bin/gnome-software --ga
user        2127    1810  0 17:32 ?        00:00:00 /usr/libexec/evolution-data-
user        2211    1684  0 17:32 ?        00:00:00 /usr/libexec/gsd-printer
user        2221    1684  0 17:32 ?        00:00:00 /usr/libexec/gvfs-udisks2-vo
user        2231    1684  0 17:32 ?        00:00:00 /usr/libexec/gvfs-mtp-volume
user        2240    1684  0 17:32 ?        00:00:00 /usr/libexec/gvfs-gphoto2-vo
user        2246    1684  0 17:32 ?        00:00:00 /usr/libexec/gvfs-goa-volume
user        2252    1684  0 17:32 ?        00:00:00 /usr/libexec/goa-daemon
user        2260    1684  0 17:32 ?        00:00:00 /usr/libexec/goa-identity-se
user        2262    1684  0 17:32 ?        00:00:00 /usr/libexec/gvfs-afc-volume
user        2273    1867  0 17:32 ?        00:00:00 /usr/bin/Xwayland :0 -rootle
user        2337    1684  0 17:32 ?        00:00:00 /usr/libexec/evolution-calen
user        2348    1684  0 17:32 ?        00:00:00 /usr/libexec/xdg-desktop-por
user        2353    1684  0 17:32 ?        00:00:00 /usr/libexec/gvfsd-metadata
user        2360    1684  0 17:32 ?        00:00:00 /usr/libexec/evolution-addre
user        2368    1684  0 17:32 ?        00:00:00 /usr/libexec/xdg-document-po
root        2377    2368  0 17:32 ?        00:00:00 fusermount3 -o rw,nosuid,nod
user        2412    1684  0 17:32 ?        00:00:00 /usr/libexec/gsd-xsettings
root        2424       2  0 17:32 ?        00:00:00 [kworker/u69:1-ttm]
user        2430    1867  0 17:32 ?        00:00:00 /usr/libexec/mutter-x11-fram
user        2496    1984  0 17:32 ?        00:00:00 /usr/libexec/ibus-dconf
user        2497    1984  0 17:32 ?        00:00:01 /usr/libexec/ibus-extension-
user        2504    1684  0 17:32 ?        00:00:00 /usr/libexec/ibus-portal
user        2506    1684  0 17:32 ?        00:00:00 /usr/libexec/localsearch-3
user        2507    1684  0 17:32 ?        00:00:00 /usr/libexec/xdg-desktop-por
user        2525    1984  0 17:32 ?        00:00:00 /usr/lib/ibus-mozc/ibus-engi
user        2533    2525  0 17:32 ?        00:00:00 /usr/lib/mozc/mozc_server
root        2548       2  0 17:32 ?        00:00:00 [kworker/u69:2-ttm]
root        2549       2  0 17:32 ?        00:00:00 [kworker/u69:3-ttm]
user        2550    1684  0 17:32 ?        00:00:00 /usr/libexec/ibus-x11
root        2563       2  0 17:32 ?        00:00:00 [kworker/u68:3-ttm]
root        2564       2  0 17:32 ?        00:00:00 [kworker/u68:4-ttm]
user        2587    1684  0 17:32 ?        00:00:00 /usr/libexec/xdg-desktop-por
root        2611       2  0 17:32 ?        00:00:00 [kworker/u68:7-ttm]
root        2613       2  0 17:32 ?        00:00:00 [kworker/u68:9-ttm]
root        2657       2  0 17:32 ?        00:00:00 [kworker/u64:3-dm_vblank_con
root        2705       2  0 17:32 ?        00:00:00 [kworker/10:3-cgroup_release
root        2707       1  0 17:32 ?        00:00:01 /usr/libexec/fwupd/fwupd
user        2766    1684  0 17:32 ?        00:00:01 /usr/libexec/gnome-terminal-
user        2775    2766  0 17:32 pts/0    00:00:00 bash
root        2818       1  0 17:32 ?        00:00:00 /usr/libexec/power-profiles-
```



```
root        2818       1  0 17:32 ?        00:00:00 /usr/libexec/power-profiles-
user        2873    1867  1 17:36 ?        00:00:03 /usr/lib/libreoffice/program
user        2907    2873 11 17:36 ?        00:00:26 /usr/lib/libreoffice/program
root        2976       2  0 17:37 ?        00:00:00 [kworker/u65:6-events_unboun
root        2978       2  0 17:37 ?        00:00:00 [kworker/0:0]
root        2979       2  0 17:38 ?        00:00:00 [kworker/2:0-events]
root        2981       2  0 17:38 ?        00:00:00 [kworker/6:0-events]
root        2983       2  0 17:38 ?        00:00:00 [kworker/u69:4-ttm]
root        2984       2  0 17:38 ?        00:00:00 [kworker/u69:5]
root        2999       2  0 17:40 ?        00:00:00 [kworker/12:0-events]
root        3001       2  0 17:40 ?        00:00:00 [kworker/14:0]
user        3002    2775  0 17:40 pts/0    00:00:00 ps -ef
user@debian:~$
```

Para realizar un filtrado de procesos se utilizó:

ps -ef | grep firefox



```
user@debian:~$ ps -ef > procesos.txt
user@debian:~$ grep firefox procesos.txt
user        3128    1867 11 17:49 ?        00:00:50 /usr/lib/firefox-esr/firefox-esr
user        3136    1684  0 17:49 ?        00:00:00 /usr/lib/firefox-esr/crashhelper 3128 9 /tmp/ 11
user        3215    3128  0 17:49 ?        00:00:00 /usr/lib/firefox-esr/firefox-esr -contentproc -parentBuildID 20260609153453 -prefsHandle 0:26640 -prefMapHandle 1:268799 -sandboxReporter 2 -chrootClient 3 -ip
cHandle 4 -initialChannelId {c9a9d146-26d9-4770-8a8f-c7b40917af4b} -parentPid 3128 -crashReporter 5 -crashHelper 6 -appDir /usr/lib/firefox-esr/browser 1 socket
user        3284    3128  0 17:50 ?        00:00:01 /usr/lib/firefox-esr/firefox-esr -contentproc -isForBrowser -prefsHandle 0:28015 -prefMapHandle 1:268799 -jsInitHandle 2:242716 -parentBuildID 20260609153453 -
sandboxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {9eba03ed-10d8-4e59-8269-40818c3dcd36} -parentPid 3128 -crashReporter 6 -crashHelper 7 -greomni /usr/lib/firefox-esr/omni.ja -appomni /usr/lib/fir
efox-esr/browser/omni.ja -appDir /usr/lib/firefox-esr/browser 2 tab
user        3292    3128  0 17:50 ?        00:00:00 /usr/lib/firefox-esr/firefox-esr -contentproc -parentBuildID 20260609153453 -prefsHandle 0:28060 -prefMapHandle 1:268799 -sandboxReporter 2 -chrootClient 3 -ip
cHandle 4 -initialChannelId {6acdca0-cb3f-4b9d-a5fd-15672d2efac4} -parentPid 3128 -crashReporter 5 -crashHelper 6 -appDir /usr/lib/firefox-esr/browser 3 rdd
user        3324    3128  0 17:50 ?        00:00:02 /usr/lib/firefox-esr/firefox-esr -contentproc -isForBrowser -prefsHandle 0:28210 -prefMapHandle 1:268799 -jsInitHandle 2:242716 -parentBuildID 20260609153453 -
sandboxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {b8a217ad-e98b-430d-94a9-f12f6965b9c9} -parentPid 3128 -crashReporter 6 -crashHelper 7 -greomni /usr/lib/firefox-esr/omni.ja -appomni /usr/lib/fir
efox-esr/browser/omni.ja -appDir /usr/lib/firefox-esr/browser 4 tab
user        3432    3128  0 17:50 ?        00:00:00 /usr/lib/firefox-esr/firefox-esr -contentproc -isForBrowser -prefsHandle 0:37626 -prefMapHandle 1:268799 -jsInitHandle 2:242716 -parentBuildID 20260609153453 -
sandboxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {f538f998-4824-4481-aee0-2a50b6145f4f} -parentPid 3128 -crashReporter 6 -crashHelper 7 -greomni /usr/lib/firefox-esr/omni.ja -appomni /usr/lib/fir
efox-esr/browser/omni.ja -appDir /usr/lib/firefox-esr/browser 6 tab
user        3457    3128  0 17:50 ?        00:00:00 /usr/lib/firefox-esr/firefox-esr -contentproc -parentBuildID 20260609153453 -sandboxingKind 0 -prefsHandle 0:40107 -prefMapHandle 1:268799 -sandboxReporter 2 -
chrootClient 3 -ipcHandle 4 -initialChannelId {12f7e58b-a7bc-4d28-91e4-dfde731c0661} -parentPid 3128 -crashReporter 5 -crashHelper 6 -appDir /usr/lib/firefox-esr/browser 7 utility
user        3466    3128  0 17:50 ?        00:00:00 /usr/lib/firefox-esr/firefox-esr -contentproc -isForBrowser -prefsHandle 0:38247 -prefMapHandle 1:268799 -jsInitHandle 2:242716 -parentBuildID 20260609153453 -
sandboxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {8ad9ee42-bc3b-4343-9c80-6e9e508ad09f} -parentPid 3128 -crashReporter 6 -crashHelper 7 -greomni /usr/lib/firefox-esr/omni.ja -appomni /usr/lib/fir
efox-esr/browser/omni.ja -appDir /usr/lib/firefox-esr/browser 8 tab
user        3470    3128  0 17:50 ?        00:00:00 /usr/lib/firefox-esr/firefox-esr -contentproc -isForBrowser -prefsHandle 0:38247 -prefMapHandle 1:268799 -jsInitHandle 2:242716 -parentBuildID 20260609153453 -
sandboxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {389059fe-8a14-4e61-976c-3165b4172cba} -parentPid 3128 -crashReporter 6 -crashHelper 7 -greomni /usr/lib/firefox-esr/omni.ja -appomni /usr/lib/fir
efox-esr/browser/omni.ja -appDir /usr/lib/firefox-esr/browser 9 tab
user        3554    3128  0 17:50 ?        00:00:00 /usr/lib/firefox-esr/firefox-esr -contentproc -isForBrowser -prefsHandle 0:38333 -prefMapHandle 1:268799 -jsInitHandle 2:242716 -parentBuildID 20260609153453 -
sandboxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {c2754b22-5627-4ed2-ae90-4441caf68ba5} -parentPid 3128 -crashReporter 6 -crashHelper 7 -greomni /usr/lib/firefox-esr/omni.ja -appomni /usr/lib/fir
efox-esr/browser/omni.ja -appDir /usr/lib/firefox-esr/browser 10 tab
user@debian:~$
```

Este comando utiliza grep para buscar dentro de la lista de procesos aquellos que contienen el término firefox.

Resultado: El comando permitió realizar una búsqueda específica dentro de los procesos mostrados por ps, facilitando la localización de procesos relacionados con Firefox.

Para consultar los procesos pertenecientes a un usuario específico se utilizó:

ps -u nombre_usuario

En este comando, nombre_usuario se sustituye por el nombre del usuario que se desea consultar.

```
user@debian:~$ ps -u user
    PID TTY          TIME CMD
   1684 ?        00:00:00 systemd
   1686 ?        00:00:00 (sd-pam)
   1706 ?        00:00:00 dbus-daemon
   1709 ?        00:00:00 pipewire
   1710 ?        00:00:00 pipewire
   1711 ?        00:00:01 wireplumber
   1712 ?        00:00:00 pipewire-pulse
   1713 ?        00:00:00 mpris-proxy
   1715 ?        00:00:00 gnome-keyring-d
   1729 tty2     00:00:00 gdm-wayland-ses
   1733 tty2     00:00:00 gnome-session-b
   1787 ?        00:00:00 gcr-ssh-agent
   1788 ?        00:00:00 gnome-session-c
   1789 ?        00:00:00 ssh-agent
   1800 ?        00:00:00 gvfsd
   1807 ?        00:00:00 gvfsd-fuse
   1810 ?        00:00:00 gnome-session-b
   1844 ?        00:00:00 at-spi-bus-laun
   1861 ?        00:00:00 dbus-daemon
   1867 ?        00:02:06 gnome-shell
   1892 ?        00:00:00 xdg-permission-
   1925 ?        00:00:00 at-spi2-registr
   1960 ?        00:00:00 gnome-shell-cal
   1969 ?        00:00:00 evolution-sourc
   1972 ?        00:00:00 dconf-service
   1974 ?        00:00:00 gjs
   1984 ?        00:00:05 ibus-daemon
   1985 ?        00:00:00 gsd-a11y-settin
   1987 ?        00:00:00 gsd-color
   1988 ?        00:00:00 gsd-datetime
   1990 ?        00:00:00 gsd-housekeepin
   1993 ?        00:00:00 gsd-keyboard
   2000 ?        00:00:00 gsd-media-keys
   2002 ?        00:00:00 gsd-power
   2007 ?        00:00:00 gsd-print-notif
   2008 ?        00:00:00 gsd-rfkill
   2011 ?        00:00:00 gsd-screensaver
   2014 ?        00:00:00 gsd-sharing
   2016 ?        00:00:00 gsd-smartcard
   2017 ?        00:00:00 gsd-sound
   2018 ?        00:00:00 gsd-usb-protect
   2019 ?        00:00:00 gsd-wacom
   2078 ?        00:00:00 gjs
   2113 ?        00:00:00 gsd-disk-utilit
   2121 ?        00:00:04 gnome-software
   2127 ?        00:00:00 evolution-alarm
   2211 ?        00:00:00 gsd-printer
   2221 ?        00:00:00 gvfs-udisks2-vo
   2231 ?        00:00:00 gvfs-mtp-volume
   2240 ?        00:00:00 gvfs-gphoto2-vo
   2246 ?        00:00:00 gvfs-goa-volume
   2252 ?        00:00:00 goa-daemon
```

```
   2252 ?        00:00:00 goa-daemon
   2260 ?        00:00:00 goa-identity-se
   2262 ?        00:00:00 gvfs-afc-volume
   2273 ?        00:00:00 Xwayland
   2337 ?        00:00:00 evolution-calen
   2348 ?        00:00:00 xdg-desktop-por
   2353 ?        00:00:00 gvfsd-metadata
   2360 ?        00:00:00 evolution-addre
   2368 ?        00:00:00 xdg-document-po
   2412 ?        00:00:00 gsd-xsettings
   2430 ?        00:00:00 mutter-x11-fram
   2496 ?        00:00:00 ibus-dconf
   2497 ?        00:00:08 ibus-extension-
   2504 ?        00:00:00 ibus-portal
   2506 ?        00:00:01 localsearch-3
   2507 ?        00:00:00 xdg-desktop-por
   2525 ?        00:00:01 ibus-engine-moz
   2533 ?        00:00:00 mozc_server
   2550 ?        00:00:00 ibus-x11
   2587 ?        00:00:00 xdg-desktop-por
   2766 ?        00:00:19 gnome-terminal-
   2775 pts/0    00:00:00 bash
   2873 ?        00:00:03 oosplash
   2907 ?        00:01:27 soffice.bin
   3128 ?        00:00:51 firefox-esr
   3136 ?        00:00:00 crashhelper
   3215 ?        00:00:00 Socket Process
   3284 ?        00:00:01 WebExtensions
   3292 ?        00:00:00 RDD Process
   3324 ?        00:00:02 Privileged Cont
   3432 ?        00:00:00 Isolated Web Co
   3457 ?        00:00:00 Utility Process
   3466 ?        00:00:00 Web Content
   3470 ?        00:00:00 Web Content
   3554 ?        00:00:00 Web Content
   3771 pts/0    00:00:00 ps
user@debian:~$
```

Resultado: Se mostraron los procesos asociados al usuario seleccionado, permitiendo identificar las tareas que estaban siendo ejecutadas por dicho usuario.

Mostrar solamente PID y comando

Después se ejecutó:

ps -eo pid,comm

Este comando permite mostrar únicamente dos datos principales: el PID y el comando correspondiente a cada proceso.

```
user@debian:~$ ps -eo pid,comm
    PID COMMAND
      1 systemd
      2 kthreadd
      3 pool_workqueue_release
      4 kworker/R-kvfree_rcu_reclaim
      5 kworker/R-rcu_gp
      6 kworker/R-sync_wq
      7 kworker/R-slub_flushwq
      8 kworker/R-netns
     11 kworker/0:0H-events_highpri
     12 kworker/u64:0-dm_vblank_control_workqueue
     13 kworker/R-mm_percpu_wq
     14 rcu_tasks_kthread
     15 rcu_tasks_rude_kthread
     16 rcu_tasks_trace_kthread
     17 ksoftirqd/0
```

```
18 rcu_preempt
19 rcu_exp_par_gp_kthread_worker/0
20 rcu_exp_gp_kthread_worker
21 migration/0
22 idle_inject/0
23 cpuhp/0
24 cpuhp/2
25 idle_inject/2
26 migration/2
27 ksoftirqd/2
29 kworker/2:0H-events_highpri
30 cpuhp/4
31 idle_inject/4
32 migration/4
33 ksoftirqd/4
35 kworker/4:0H-events_highpri
36 cpuhp/6
37 idle_inject/6
38 migration/6
39 ksoftirqd/6
41 kworker/6:0H-events_highpri
42 cpuhp/8
43 idle_inject/8
44 migration/8
45 ksoftirqd/8
47 kworker/8:0H-events_highpri
48 cpuhp/10
49 idle_inject/10
50 migration/10
51 ksoftirqd/10
53 kworker/10:0H-events_highpri
54 cpuhp/12
55 idle_inject/12
56 migration/12
57 ksoftirqd/12
59 kworker/12:0H-events_highpri
60 cpuhp/14
61 idle_inject/14
62 migration/14
63 ksoftirqd/14
65 kworker/14:0H-events_highpri
66 cpuhp/1
67 idle_inject/1
68 migration/1
69 ksoftirqd/1
71 kworker/1:0H-events_highpri
72 cpuhp/3
73 idle_inject/3
74 migration/3
75 ksoftirqd/3
76 kworker/3:0-events
77 kworker/3:0H-events_highpri
78 cpuhp/5
79 idle_inject/5
80 migration/5
81 ksoftirqd/5
82 kworker/5:0-events
83 kworker/5:0H-events_highpri
84 cpuhp/7
85 idle_inject/7
86 migration/7
87 ksoftirqd/7
89 kworker/7:0H-events_highpri
90 cpuhp/9
91 idle_inject/9
92 migration/9
93 ksoftirqd/9
95 kworker/9:0H-events_highpri
96 cpuhp/11
97 idle_inject/11
98 migration/11
99 ksoftirqd/11
100 kworker/11:0-events
101 kworker/11:0H-events_highpri
102 cpuhp/13
103 idle_inject/13
104 migration/13
105 ksoftirqd/13
107 kworker/13:0H-events_highpri
108 cpuhp/15
109 idle_inject/15
110 migration/15
111 ksoftirqd/15
113 kworker/15:0H-events_highpri
120 kworker/u66:4-events_unbound
125 kworker/u66:9-kvfree_rcu_reclaim
126 kworker/u66:10-kvfree_rcu_reclaim
128 kdevtmpfs
129 kworker/R-inet_frag_wq
130 kauditd
131 khungtaskd
133 oom_reaper
135 kworker/R-writeback
136 kcompactd0
137 ksmd
138 khugepaged
139 kworker/R-kintegrityd
140 kworker/R-kblockd
141 kworker/R-blkcg_punt_bio
142 irq/9-acpi
143 kworker/1:1-events
144 kworker/2:1-events
145 kworker/R-tpm_dev_wq
146 kworker/R-edac-poller
148 kworker/R-devfreq_wq
149 kworker/11:1-events
150 kworker/R-quota_events_unbound
150 kworker/R-quota_events_unbound
151 kworker/1:1H-kblockd
152 irq/25-AMD-Vi0-Evt
153 irq/26-AMD-Vi0-PPR
154 irq/27-AMD-Vi0-GA
155 kswapd0
157 kworker/10:1-events
160 kworker/7:1-events
166 kworker/R-kthrotld
170 kworker/R-acpi_thermal_pm
171 kworker/12:1-events
173 kworker/R-mld
174 kworker/9:1H-kblockd
175 kworker/R-ipv6_addrconf
180 kworker/R-kstrp
182 kworker/u67:0-hci0
236 kworker/0:1H-kblockd
250 kworker/3:1-mm_percpu_wq
251 kworker/8:1-events
257 kworker/15:1-events
274 kworker/12:1H-kblockd
283 kworker/2:1H-kblockd
295 kworker/11:1H-kblockd
300 kworker/4:1H-kblockd
302 kworker/6:1H-kblockd
316 kworker/15:1H-kblockd
318 kworker/13:1H-kblockd
319 kworker/14:1H-kblockd
320 kworker/8:1H-kblockd
321 kworker/7:1H-kblockd
322 kworker/5:1H-kblockd
323 kworker/10:1H-kblockd
324 kworker/3:1H-kblockd
327 kworker/13:2-mm_percpu_wq
342 watchdogd
343 kworker/R-nvme-wq
344 kworker/R-nvme-reset-wq
345 kworker/R-nvme-delete-wq
346 kworker/R-nvme-auth-wq
347 irq/33-ELAN06FA:00
348 kworker/R-sdhci
349 irq/35-mmc0
351 kworker/R-ata_sff
353 scsi_eh_0
354 kworker/R-scsi_tmf_0
355 scsi_eh_1
356 kworker/R-scsi_tmf_1
357 kworker/u65:3-kvfree_rcu_reclaim
364 scsi_eh_2
365 kworker/R-scsi_tmf_2
366 usb-storage
367 kworker/R-uas
369 kworker/R-amdgpu-reset-dev
370 kworker/R-ttm
371 kworker/R-amdgpu_dm_hpd_rx_offload_wq
```

```
 372 kworker/R-amdgpu_dm_hpd_rx_offload_wq
 373 kworker/R-amdgpu_dm_hpd_rx_offload_wq
 374 kworker/R-dm_vblank_control_workqueue
 375 card0-crtc0
 376 card0-crtc1
 377 card0-crtc2
 378 card0-crtc3
 379 kworker/R-gfx
 380 kworker/R-comp_1.0.0
 381 kworker/R-comp_1.1.0
 382 kworker/R-comp_1.2.0
 383 kworker/R-comp_1.3.0
 384 kworker/R-comp_1.0.1
 385 kworker/R-comp_1.1.1
 386 kworker/R-comp_1.2.1
 387 kworker/R-comp_1.3.1
 388 kworker/R-sdma0
 389 kworker/R-vcn_dec
 390 kworker/R-vcn_enc0
 391 kworker/R-vcn_enc1
 392 kworker/R-jpeg_dec
 401 kworker/R-md
 402 kworker/R-md_bitmap
 408 kworker/R-raid5wq
 539 kworker/14:2-cgroup_free
 589 psimon
 603 systemd-journal
 659 systemd-udevd
 661 psimon
 750 kworker/R-cryptd
 834 kworker/R-cfg80211
 838 kworker/9:3-events
 926 kworker/u67:2-hci0
 936 napi/phy0-8193
 937 napi/phy0-8194
 938 napi/phy0-8195
1197 mt76-tx phy0
1539 accounts-daemon
1541 avahi-daemon
1542 bluetoothd
1543 cron
1548 dbus-daemon
1551 low-memory-moni
1552 polkitd
1553 smartd
1554 switcheroo-cont
1556 systemd-logind
1557 udisksd
1558 avahi-daemon
1592 NetworkManager
1593 wpa_supplicant
1599 psimon
1626 ModemManager
1636 krfcommd
1652 cupsd
1670 gdm3
1676 gdm-session-wor
1684 systemd
1686 (sd-pam)
1706 dbus-daemon
1709 pipewire
1710 pipewire
1711 wireplumber
1712 pipewire-pulse
1713 mpris-proxy
1715 gnome-keyring-d
1724 rtkit-daemon
1729 gdm-wayland-ses
1733 gnome-session-b
1787 gcr-ssh-agent
1788 gnome-session-c
1789 ssh-agent
1800 gvfsd
1807 gvfsd-fuse
1810 gnome-session-b
1844 at-spi-bus-laun
1861 dbus-daemon
1867 gnome-shell
1888 cups-browsed
1892 xdg-permission-
1923 kworker/u64:2-comp_1.3.0
1925 at-spi2-registr
1926 colord
1960 gnome-shell-cal
1969 evolution-sourc
1972 dconf-service
1974 gjs
1976 upowerd
1984 ibus-daemon
1985 gsd-a11y-settin
1987 gsd-color
1988 gsd-datetime
1990 gsd-housekeepin
1993 gsd-keyboard
2000 gsd-media-keys
2002 gsd-power
2007 gsd-print-notif
2008 gsd-rfkill
2011 gsd-screensaver
2014 gsd-sharing
2016 gsd-smartcard
2017 gsd-sound
2018 gsd-usb-protect
2019 gsd-wacom
2078 gjs
2113 gsd-disk-utilit
2121 gnome-software
2127 evolution-alarm
2211 gsd-printer
2211 gsd-printer
2221 gvfs-udisks2-vo
2231 gvfs-mtp-volume
2240 gvfs-gphoto2-vo
2246 gvfs-goa-volume
2252 goa-daemon
2260 goa-identity-se
2262 gvfs-afc-volume
2273 Xwayland
2337 evolution-calen
2348 xdg-desktop-por
2353 gvfsd-metadata
2360 evolution-addre
2368 xdg-document-po
2377 fusermount3
2412 gsd-xsettings
2430 mutter-x11-fram
2496 ibus-dconf
2497 ibus-extension-
2504 ibus-portal
2506 localsearch-3
2507 xdg-desktop-por
2525 ibus-engine-moz
2533 mozc_server
2550 ibus-x11
2564 kworker/u68:4-ttm
2587 xdg-desktop-por
2613 kworker/u68:9-ttm
2707 fwupd
2766 gnome-terminal-
2775 bash
2818 power-profiles-
2873 oosplash
2907 soffice.bin
2976 kworker/u65:6-kvfree_rcu_reclaim
3020 kworker/13:0
3100 kworker/2:2-events
3114 kworker/u69:5-ttm
3128 firefox-esr
3136 crashhelper
3215 Socket Process
3230 kworker/15:0-events
3284 WebExtensions
3292 RDD Process
3324 Privileged Cont
3432 Isolated Web Co
3457 Utility Process
3466 Web Content
3470 Web Content
3520 kworker/u65:0-events_unbound
3521 kworker/8:2-events
3554 Web Content
3580 kworker/0:0-events
3599 kworker/7:0
3600 kworker/1:2
```

```
   3612 kworker/4:1-events_freezable_pwr_efficient
   3677 kworker/9:0-events
   3682 kworker/6:0-events
   3685 kworker/14:0-events
   3687 kworker/5:2-events_freezable_pwr_efficient
   3692 kworker/10:2-events
   3707 kworker/0:1-events
   3728 kworker/u64:1-comp_1.3.0
   3730 kworker/u69:1-ttm
   3733 kworker/u69:2-ttm
   3734 kworker/u69:3-ttm
   3735 kworker/u69:4
   3738 kworker/u68:0-ttm
   3739 kworker/u68:1-ttm
   3749 kworker/u69:6-ttm
   3750 kworker/4:2-events
   3751 kworker/6:1-events
   3752 kworker/12:2-mm_percpu_wq
   3761 kworker/14:1
   3767 kworker/10:0-events
   3768 kworker/10:3
   3775 kworker/u68:2-ttm
   3776 kworker/u68:3
   3779 kworker/3:2-mm_percpu_wq
   3788 kworker/5:1-events_freezable
   3789 kworker/u65:1-kvfree_rcu_reclaim
   3792 kworker/8:0-events
   3809 kworker/0:2-mm_percpu_wq
   3810 ps
user@debian:~$
```

Resultado: Se obtuvo una lista simplificada de procesos, mostrando su identificador de proceso (PID) y el nombre del comando que corresponde a cada uno.

Ordenar procesos por uso de CPU

Finalmente se ejecutó:

ps -eo pid,comm,%cpu --sort=-%cpu

Este comando muestra el PID, el nombre del comando y el porcentaje de CPU utilizado. Además, los procesos se ordenan de acuerdo con su consumo de CPU.

```
user@debian:~$ ps -eo pid,comm,%cpu --sort=-%cpu
    PID COMMAND         %CPU
   1867 gnome-shell      6.9
   2907 soffice.bin      4.5
   4042 gnome-control-c  2.7
   3128 firefox-esr      2.3
   2766 gnome-terminal-  1.0
    347 irq/33-ELAN06FA  0.3
   1984 ibus-daemon      0.2
   2497 ibus-extension-  0.2
   3324 Privileged Cont  0.2
   2121 gnome-software   0.1
   3470 Web Content      0.1
   3554 Web Content      0.1
   2873 oosplash         0.1
```

```
4133 ibus-engine-sim  0.0
2525 ibus-engine-moz  0.0
3520 kworker/u65:0-e  0.0
  18 rcu_preempt      0.0
2506 localsearch-3    0.0
3789 kworker/u65:1-k  0.0
1711 wireplumber      0.0
   1 systemd          0.0
3284 WebExtensions    0.0
2707 fwupd            0.0
2976 kworker/u65:6-k  0.0
1709 pipewire         0.0
 366 usb-storage      0.0
1923 kworker/u64:2-s  0.0
3432 Isolated Web Co  0.0
3976 kworker/u64:1-g  0.0
1548 dbus-daemon      0.0
2127 evolution-alarm  0.0
1712 pipewire-pulse   0.0
2273 Xwayland         0.0
 120 kworker/u66:4-k  0.0
1706 dbus-daemon      0.0
2533 mozc_server      0.0
 125 kworker/u66:9-e  0.0
1969 evolution-sourc  0.0
3466 Web Content      0.0
2000 gsd-media-keys   0.0
 138 khugepaged       0.0
1552 polkitd          0.0
1976 upowerd          0.0
1592 NetworkManager   0.0
1684 systemd          0.0
 603 systemd-journal  0.0
2507 xdg-desktop-por  0.0
3292 RDD Process      0.0
2348 xdg-desktop-por  0.0
1557 udisksd          0.0
4151 kworker/6:0-eve  0.0
3809 kworker/0:2-eve  0.0
 126 kworker/u66:10-  0.0
3954 kworker/14:1-ev  0.0
2360 evolution-addre  0.0
  44 migration/8      0.0
3215 Socket Process   0.0
1926 colord           0.0
 659 systemd-udevd    0.0
  80 migration/5      0.0
  26 migration/2      0.0
2430 mutter-x11-fram  0.0
2262 gvfs-afc-volume  0.0
2002 gsd-power        0.0
  68 migration/1      0.0
  74 migration/3      0.0
  86 migration/7      0.0
  92 migration/9      0.0
  98 migration/11     0.0
 104 migration/13     0.0
 110 migration/15     0.0
  32 migration/4      0.0
  38 migration/6      0.0
  50 migration/10     0.0
  56 migration/12     0.0
  62 migration/14     0.0
1593 wpa_supplicant   0.0
 136 kcompactd0       0.0
1556 systemd-logind   0.0
1960 gnome-shell-cal  0.0
1810 gnome-session-b  0.0
1626 ModemManager     0.0
2337 evolution-calen  0.0
3811 kworker/4:0-eve  0.0
3855 kworker/8:2-cgr  0.0
2412 gsd-xsettings    0.0
1993 gsd-keyboard     0.0
1987 gsd-color        0.0
3457 Utility Process  0.0
1990 gsd-housekeepin  0.0
 257 kworker/15:1-ev  0.0
2019 gsd-wacom        0.0
1715 gnome-keyring-d  0.0
 250 kworker/3:1-eve  0.0
1539 accounts-daemon  0.0
2818 power-profiles-  0.0
1733 gnome-session-b  0.0
 160 kworker/7:1-eve  0.0
3752 kworker/12:2-ev  0.0
2587 xdg-desktop-por  0.0
2078 gjs              0.0
1542 bluetoothd       0.0
 182 kworker/u67:0-h  0.0
3990 kworker/2:1-eve  0.0
3677 kworker/9:0-eve  0.0
1724 rtkit-daemon     0.0
1553 smartd           0.0
3982 kworker/10:0-mm  0.0
4108 kworker/8:0-eve  0.0
2775 bash             0.0
  17 ksoftirqd/0      0.0
  57 ksoftirqd/12     0.0
2221 gvfs-udisks2-vo  0.0
2014 gsd-sharing      0.0
1974 gjs              0.0
1652 cupsd            0.0
 149 kworker/11:1-mm  0.0
  63 ksoftirqd/14     0.0
4013 kworker/6:2-eve  0.0
1800 gvfsd            0.0
2016 gsd-smartcard    0.0
2017 gsd-sound        0.0
1670 gdm3             0.0
1541 avahi-daemon     0.0
 316 kworker/15:1H-k  0.0
 143 kworker/1:1-eve  0.0
 111 ksoftirqd/15     0.0
  51 ksoftirqd/10     0.0
3687 kworker/5:2-eve  0.0
2252 goa-daemon       0.0
1888 cups-browsed     0.0
1713 mpris-proxy      0.0
1676 gdm-session-wor  0.0
1551 low-memory-moni  0.0
 327 kworker/13:2-ev  0.0
 320 kworker/8:1H-kb  0.0
 302 kworker/6:1H-kb  0.0
 236 kworker/0:1H-kb  0.0
  93 ksoftirqd/9      0.0
  99 ksoftirqd/11     0.0
 105 ksoftirqd/13     0.0
  27 ksoftirqd/2      0.0
  39 ksoftirqd/6      0.0
  45 ksoftirqd/8      0.0
3952 kworker/15:2-ev  0.0
3943 kworker/1:0-eve  0.0
2550 ibus-x11         0.0
2368 xdg-document-po  0.0
2260 goa-identity-se  0.0
2211 gsd-printer      0.0
2113 gsd-disk-utilit  0.0
2011 gsd-screensaver  0.0
2018 gsd-usb-protect  0.0
1972 dconf-service    0.0
1861 dbus-daemon      0.0
1710 pipewire         0.0
 300 kworker/4:1H-kb  0.0
  69 ksoftirqd/1      0.0
  21 migration/0      0.0
  33 ksoftirqd/4      0.0
2504 ibus-portal      0.0
2007 gsd-print-notif  0.0
2008 gsd-rfkill       0.0
1988 gsd-datetime     0.0
1985 gsd-a11y-settin  0.0
1925 at-spi2-registr  0.0
1789 ssh-agent        0.0
1554 switcheroo-cont  0.0
1543 cron             0.0
 938 napi/phy0-8195   0.0
 926 kworker/u67:2-h  0.0
 323 kworker/10:1H-k  0.0
 319 kworker/14:1H-k  0.0
 283 kworker/2:1H-kb  0.0
 274 kworker/12:1H-k  0.0
  75 ksoftirqd/3      0.0
  81 ksoftirqd/5      0.0
2564 kworker/u68:4-t  0.0
2496 ibus-dconf       0.0
2353 gvfsd-metadata   0.0
2246 gvfs-goa-volume  0.0
2240 gvfs-gphoto2-vo  0.0
2231 gvfs-mtp-volume  0.0
1892 xdg-permission-  0.0
1844 at-spi-bus-laun  0.0
1807 gvfsd-fuse       0.0
1787 gcr-ssh-agent    0.0
1729 gdm-wayland-ses  0.0
 324 kworker/3:1H-kb  0.0
 321 kworker/7:1H-kb  0.0
 322 kworker/5:1H-kb  0.0
 151 kworker/1:1H-kb  0.0
  87 ksoftirqd/7      0.0
   2 kthreadd         0.0
   3 pool_workqueue_  0.0
   4 kworker/R-kvfre  0.0
   5 kworker/R-rcu_g  0.0
   6 kworker/R-sync_  0.0
   7 kworker/R-slub_  0.0
   8 kworker/R-netns  0.0
  11 kworker/0:0H-ev  0.0
  13 kworker/R-mm_pe  0.0
  14 rcu_tasks_kthre  0.0
  15 rcu_tasks_rude_  0.0
  16 rcu_tasks_trace  0.0
  19 rcu_exp_par_gp_  0.0
  20 rcu_exp_gp_kthr  0.0
  22 idle_inject/0    0.0
  23 cpuhp/0          0.0
  24 cpuhp/2          0.0
  25 idle_inject/2    0.0
  29 kworker/2:0H-ev  0.0
  30 cpuhp/4          0.0
  31 idle_inject/4    0.0
  35 kworker/4:0H-ev  0.0
  36 cpuhp/6          0.0
  37 idle_inject/6    0.0
  41 kworker/6:0H-ev  0.0
  42 cpuhp/8          0.0
  43 idle_inject/8    0.0
  47 kworker/8:0H-ev  0.0
  48 cpuhp/10         0.0
  49 idle_inject/10   0.0
  53 kworker/10:0H-e  0.0
  54 cpuhp/12         0.0
  55 idle_inject/12   0.0
  59 kworker/12:0H-e  0.0
  60 cpuhp/14         0.0
  61 idle_inject/14   0.0
  65 kworker/14:0H-e  0.0
  66 cpuhp/1          0.0
  67 idle_inject/1    0.0
  71 kworker/1:0H-ev  0.0
  72 cpuhp/3          0.0
  73 idle_inject/3    0.0
  77 kworker/3:0H-ev  0.0
  78 cpuhp/5          0.0
  79 idle_inject/5    0.0
  83 kworker/5:0H-ev  0.0
  84 cpuhp/7          0.0
  85 idle_inject/7    0.0
  89 kworker/7:0H-ev  0.0
  90 cpuhp/9          0.0
  91 idle_inject/9    0.0
  95 kworker/9:0H-ev  0.0
  96 cpuhp/11         0.0
  97 idle_inject/11   0.0
 101 kworker/11:0H-e  0.0
 102 cpuhp/13         0.0
 103 idle_inject/13   0.0
 107 kworker/13:0H-e  0.0
 108 cpuhp/15         0.0
 109 idle_inject/15   0.0
 113 kworker/15:0H-e  0.0
 128 kdevtmpfs        0.0
 129 kworker/R-inet_  0.0
 130 kauditd          0.0
 131 khungtaskd       0.0
 133 oom_reaper       0.0
 135 kworker/R-write  0.0
 137 ksmd             0.0
 139 kworker/R-kinte  0.0
 140 kworker/R-kbloc  0.0
 141 kworker/R-blkcg  0.0
 142 irq/9-acpi       0.0
 145 kworker/R-tpm_d  0.0
 146 kworker/R-edac-  0.0
 148 kworker/R-devfr  0.0
 150 kworker/R-quota  0.0
 152 irq/25-AMD-Vi0-  0.0
```

```
154 irq/27-AMD-Vi0-   0.0
155 kswapd0           0.0
166 kworker/R-kthro   0.0
170 kworker/R-acpi_   0.0
173 kworker/R-mld     0.0
174 kworker/9:1H-kb   0.0
175 kworker/R-ipv6_   0.0
180 kworker/R-kstrp   0.0
295 kworker/11:1H-k   0.0
318 kworker/13:1H-k   0.0
342 watchdogd         0.0
343 kworker/R-nvme-   0.0
344 kworker/R-nvme-   0.0
345 kworker/R-nvme-   0.0
346 kworker/R-nvme-   0.0
348 kworker/R-sdhci   0.0
349 irq/35-mmc0       0.0
351 kworker/R-ata_s   0.0
353 scsi_eh_0         0.0
354 kworker/R-scsi_   0.0
355 scsi_eh_1         0.0
356 kworker/R-scsi_   0.0
364 scsi_eh_2         0.0
365 kworker/R-scsi_   0.0
367 kworker/R-uas     0.0
369 kworker/R-amdgp   0.0
370 kworker/R-ttm     0.0
371 kworker/R-amdgp   0.0
372 kworker/R-amdgp   0.0
373 kworker/R-amdgp   0.0
374 kworker/R-dm_vb   0.0
375 card0-crtc0       0.0
376 card0-crtc1       0.0
377 card0-crtc2       0.0
378 card0-crtc3       0.0
379 kworker/R-gfx     0.0
380 kworker/R-comp_   0.0
381 kworker/R-comp_   0.0
382 kworker/R-comp_   0.0
383 kworker/R-comp_   0.0
384 kworker/R-comp_   0.0
385 kworker/R-comp_   0.0
386 kworker/R-comp_   0.0
387 kworker/R-comp_   0.0
388 kworker/R-sdma0   0.0
389 kworker/R-vcn_d   0.0
390 kworker/R-vcn_e   0.0
391 kworker/R-vcn_e   0.0
392 kworker/R-jpeg_   0.0
401 kworker/R-md      0.0
402 kworker/R-md_bi   0.0
408 kworker/R-raid5   0.0
589 psimon            0.0
661 psimon            0.0
750 kworker/R-crypt   0.0
 834 kworker/R-cfg80  0.0
 936 napi/phy0-8193   0.0
 937 napi/phy0-8194   0.0
1197 mt76-tx phy0     0.0
1558 avahi-daemon     0.0
1599 psimon           0.0
1636 krfcommd         0.0
1686 (sd-pam)         0.0
1788 gnome-session-c  0.0
2377 fusermount3      0.0
3020 kworker/13:0     0.0
3100 kworker/2:2-mm_  0.0
3136 crashhelper      0.0
3692 kworker/10:2-ev  0.0
3749 kworker/u69:6-t  0.0
3751 kworker/6:1-eve  0.0
3788 kworker/5:1-eve  0.0
3925 kworker/3:0-eve  0.0
3981 kworker/9:2      0.0
3986 kworker/4:2-cgr  0.0
3987 kworker/0:1-eve  0.0
3989 kworker/12:1-ev  0.0
4006 kworker/14:2-ev  0.0
4101 kworker/u68:2-t  0.0
4107 kworker/u69:0-t  0.0
4115 kworker/11:2     0.0
4129 kworker/7:0      0.0
4144 kworker/14:0-ev  0.0
4145 kworker/15:0-ev  0.0
4149 kworker/1:2-eve  0.0
4168 kworker/8:1-mm_  0.0
4170 kworker/2:0-eve  0.0
4174 ps               0.0
er@debian:~$
```

Resultado: Se obtuvo una lista de procesos acompañada de su porcentaje de utilización de CPU. La información se presentó ordenada para facilitar la identificación de los procesos que utilizan una mayor cantidad de recursos del procesador.

Conclusiones

Durante esta práctica se aprendió a utilizar diferentes opciones del comando ps en Linux para consultar y analizar los procesos activos del sistema.

Se comprobó que ps permite obtener información de los procesos de diferentes maneras, desde una consulta básica hasta información más detallada. También se aprendió a filtrar procesos específicos, consultar los procesos de un usuario, mostrar únicamente el PID y el comando, y ordenar los procesos de acuerdo con su consumo de CPU.

La práctica permitió comprender de manera más clara cómo se pueden consultar y analizar los procesos que se ejecutan en un sistema Linux.